

Practical 6:

```
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ nano server.c
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ nano client.c
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ gcc server.c -o server
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ gcc client.c -o client
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ ls
client  client.c  server  server.c
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ ./server
Server started...
Waiting for client message...
Client: Hello
Server: Server processed: Hello

Waiting for client message...
Client: Hi Hello
Server: Server processed: Hi Hello

Waiting for client message...
Client: Hello Linux
Server: Server processed: Hello Linux

Waiting for client message...
Client: exit
Server: Server shutting down.
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ |
```

```
bhanu@DESKTOP-DE32HQ8:~$ cd ~/fifo_project/fifo_project
./client
Enter message: Hello Linux
Server: Server processed: Hello Linux
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ |
```

```
bhanu@DESKTOP-DE32HQ8:~$ cd ~/fifo_project/fifo_project
ls -l
total 44
-rwxr-xr-x 1 bhanu bhanu 16352 Sep 15 06:42 client
-rw-r--r-- 1 bhanu bhanu 790 Sep 15 06:41 client.c
prw-r--r-- 1 bhanu bhanu 0 Sep 15 06:43 client_to_server
-rwxr-xr-x 1 bhanu bhanu 16416 Sep 15 06:42 server
-rw-r--r-- 1 bhanu bhanu 1568 Sep 15 06:41 server.c
prw-r--r-- 1 bhanu bhanu 0 Sep 15 06:43 server_to_client
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ cd ~/fifo_project/fifo_project
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ ./client
Enter message: Hi Hello
Server: Server processed: Hi Hello
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ |
```

```
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ nano server.c
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ nano client.c
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ gcc server.c -o server
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ gcc client.c -o client
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ ls
client  client.c  server  server.c
bhanu@DESKTOP-DE32HQ8:~/fifo_project/fifo_project$ ./server
Server started...
Waiting for client message...
Client: Hello
Server: Server processed: Hello

Waiting for client message...
```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>

#define FIF01 "client_to_server"
#define FIF02 "server_to_client"

int main() {
    char message[100];
    char response[150];

    printf("Enter message: ");
    fgets(message, sizeof(message), stdin);

    message[strcspn(message, "\n")] = '\0';

    // Open FIFO for writing
    int fd1 = open(FIF01, O_WRONLY);

    // Send message to server
    write(fd1, message, strlen(message) + 1);

    close(fd1);

    // Open FIFO for reading
    int fd2 = open(FIF02, O_RDONLY);

    // Receive server response
    int n = read(fd2, response, sizeof(response) - 1);

    if (n > 0) {
        response[n] = '\0';
        printf("Server: %s\n", response);
    }
}
```

```
    if (strcmp(buffer, "hello") == 0) {
        strcpy(response, "Hello Client! Message received.");
    }
    else if (strcmp(buffer, "exit") == 0) {
        strcpy(response, "Server shutting down.");
    }
    else {
        snprintf(response, sizeof(response),
            "Server processed: %s", buffer);
    }

    close(fd1);

    // Open FIFO for writing
    int fd2 = open(FIFO2, O_WRONLY);

    write(fd2, response, strlen(response) + 1);

    close(fd2);

    printf("Server: %s\n", response);

    if (strcmp(buffer, "exit") == 0) {
        break;
    }

    printf("\nWaiting for client message...\n");
}
}

unlink(FIFO1);
unlink(FIFO2);

return 0;
}
```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/stat.h>

#define FIFO1 "client_to_server"
#define FIFO2 "server_to_client"

int main() {
    char buffer[100];

    // Create FIFOs
    mkfifo(FIFO1, 0666);
    mkfifo(FIFO2, 0666);

    printf("Server started...\n");
    printf("Waiting for client message...\n");

    while (1) {

        // Open FIFO for reading
        int fd1 = open(FIFO1, O_RDONLY);

        // Read message
        int n = read(fd1, buffer, sizeof(buffer) - 1);

        if (n > 0) {
            buffer[n] = '\0';

            printf("Client: %s\n", buffer);

            // Process the message
            char response[150];

            if (strcmp(buffer, "hello") == 0) {
```